

Data Visualization with Matplotlib (Detailed Guide)

What is Matplotlib?

Matplotlib is the most popular Python library used for **Data Visualization**. It helps convert data into graphs, charts, and plots so that data becomes easier to understand.

Why Use Matplotlib?

- Visualize large datasets easily
 - Identify trends and patterns
 - Compare values
 - Create professional charts
 - Used in Data Science, Machine Learning, and Analytics
-

Installation

```
pip install matplotlib
```

Import Matplotlib

```
import matplotlib.pyplot as plt
```

`pyplot` is a module that provides functions similar to MATLAB.

1. Line Plot

Used to show trends over time.

Example

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [10, 20, 15, 30, 25]

plt.plot(x, y)
plt.show()
```

Output

Shows points connected by lines.

Real-Life Use

- Stock prices
- Temperature changes
- Sales growth

Line Plot with Customization

```
import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr"]
sales = [100, 150, 120, 180]

plt.plot(
    months,
    sales,
    color="red",
    linestyle="--",
    marker="o",
    linewidth=3
)

plt.title("Monthly Sales")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.grid(True)

plt.show()
```

Explanation

Parameter	Meaning
color="red"	Line color
linestyle="--"	Dashed line
marker="o"	Circle marker
linewidth=3	Thickness of line

2. Bar Chart

Used for comparing categories.

Example

```
import matplotlib.pyplot as plt

students = ["A", "B", "C", "D"]
```

```
marks = [80, 90, 75, 95]

plt.bar(students, marks)

plt.title("Student Marks")
plt.xlabel("Students")
plt.ylabel("Marks")

plt.show()
```

Real-Life Use

- Student marks
 - Product sales
 - Employee performance
-

3. Horizontal Bar Chart

Used when category names are long.

Example

```
import matplotlib.pyplot as plt

subjects = ["Python", "Java", "PHP", "C"]
students = [50, 40, 35, 20]

plt.barh(subjects, students)

plt.title("Students Learning Languages")

plt.show()
```

Output

Horizontal bars instead of vertical bars.

4. Pie Chart

Shows percentage distribution.

Example

```
import matplotlib.pyplot as plt

sizes = [40, 30, 20, 10]
labels = ["Python", "Java", "PHP", "C"]
```

```
plt.pie(
    sizes,
    labels=labels,
    autopct="%1.1f%%"
)

plt.title("Programming Language Usage")

plt.show()
```

Explanation

Parameter	Meaning
labels	Labels for slices
autopct	Show percentages

Real-Life Use

- Market Share
 - Budget Distribution
 - Survey Results
-

Pie Chart with Explode

Used to highlight a specific slice.

```
import matplotlib.pyplot as plt

sizes = [40, 30, 20, 10]
labels = ["Python", "Java", "PHP", "C"]

explode = [0.1, 0, 0, 0]

plt.pie(
    sizes,
    labels=labels,
    explode=explode,
    autopct="%1.1f%%"
)

plt.title("Programming Language Usage")

plt.show()
```

Explanation

```
explode = [0.1, 0, 0, 0]
```

- First slice moves outward.
- 0 means no movement.
- Higher value = more separation.

5. Histogram

Shows frequency distribution.

Example

```
import matplotlib.pyplot as plt

marks = [45,50,55,60,65,70,75,80,85,90]

plt.hist(marks, bins=5)

plt.title("Marks Distribution")

plt.show()
```

Explanation

bins=5

Divides data into 5 groups.

Real-Life Use

- Exam score distribution
- Age distribution
- Salary analysis

6. Scatter Plot

Shows relationship between two variables.

Example

```
import matplotlib.pyplot as plt

height = [150,155,160,165,170]
weight = [50,55,58,65,70]

plt.scatter(height, weight)

plt.xlabel("Height")
plt.ylabel("Weight")

plt.show()
```

Real-Life Use

- Height vs Weight
- Advertising vs Sales
- Experience vs Salary

7. Subplots

Display multiple charts in one figure.

Example

```
import matplotlib.pyplot as plt

x = [1,2,3,4]

plt.subplot(1,2,1)
plt.plot(x,[1,4,9,16])
plt.title("Plot 1")

plt.subplot(1,2,2)
plt.plot(x,[1,2,3,4])
plt.title("Plot 2")

plt.show()
```

Understanding subplot()

```
plt.subplot(rows, columns, position)
```

Example:

```
plt.subplot(1,2,1)
```

Means:

Value	Meaning
1	Total rows
2	Total columns
1	First position

Layout:

```
+-----+-----+
| Plot 1 | Plot 2 |
+-----+-----+
```

8. Grid

Adds reference lines to charts.

Example

```
import matplotlib.pyplot as plt

x = [1,2,3,4]
y = [10,20,30,40]

plt.plot(x,y)

plt.grid(True)

plt.show()
```

9. Legends

Used to identify multiple datasets.

Example

```
import matplotlib.pyplot as plt

x = [1,2,3,4]

plt.plot(x, [1,4,9,16], label="Square")
plt.plot(x, [1,2,3,4], label="Linear")

plt.legend()

plt.show()
```

Output

Legend box appears showing:

```
Square
Linear
```

10. Figure Size

Controls graph size.

Example

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))

plt.plot([1,2,3], [10,20,30])
```

```
plt.show()
```

Understanding figsize

```
figsize=(8,5)
```

Means:

Value	Meaning
8	Width in inches
5	Height in inches

11. Save Plot as Image

```
import matplotlib.pyplot as plt

plt.plot([1,2,3],[10,20,30])

plt.savefig("graph.png")

plt.show()
```

Output

Creates:

```
graph.png
```

in current working directory.

Common Plot Styles

Property	Example
Color	color="red"
Line Style	linestyle="--"
Marker	marker="o"
Width	linewidth=2
Title	plt.title()
X Label	plt.xlabel()
Y Label	plt.ylabel()
Grid	plt.grid()
Legend	plt.legend()

Example: Student Marks Visualization

Data

Student Marks

Krish	85
Raj	90
Jay	78
Meet	95

Code

```
import matplotlib.pyplot as plt

students = ["Krish", "Raj", "Jay", "Meet"]
marks = [85, 90, 78, 95]

plt.bar(students, marks)

plt.title("Student Marks")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.grid(axis="y")

plt.show()
```

Output

Student Marks chart showing marks obtained by students.

Matplotlib Workflow

```
Import matplotlib
  ↓
Prepare Data
  ↓
Choose Plot Type
  ↓
Add Title & Labels
  ↓
Customize Plot
  ↓
Show Plot
```

Most Important Functions for Exams

```
plt.plot()      # Line Chart
```

```
plt.bar()           # Bar Chart
plt.barh()          # Horizontal Bar Chart
plt.scatter()       # Scatter Plot
plt.hist()          # Histogram
plt.pie()           # Pie Chart
plt.subplot()       # Multiple Plots
plt.title()         # Title
plt.xlabel()        # X-axis Label
plt.ylabel()        # Y-axis Label
plt.legend()        # Legend
plt.grid()          # Grid
plt.figure()        # Create Figure
plt.savefig()       # Save Image
plt.show()          # Display Plot
```

Interview Questions

Q1. Difference between Bar Chart and Histogram?

Bar Chart	Histogram
Compares categories	Shows distribution
Gaps between bars	No gaps
Categorical data	Continuous numerical data

Q2. Difference between Scatter Plot and Line Plot?

Scatter Plot	Line Plot
Shows relationship	Shows trend
Points only	Connected points
Correlation analysis	Time-series analysis

Q3. What is figsize?

Answer:

figsize is used to set the width and height of the graph.

Example:

```
plt.figure(figsize=(10,6))
```

Width = 10 inches

Height = 6 inches

Q4. What is explode in Pie Chart?

Answer:

`explode` separates one or more slices from the pie chart to highlight them.

Example:

```
explode = [0.1, 0, 0, 0]
```

Q5. Why use plt.show()?

Answer:

`plt.show()` displays the chart on the screen.

Without it, the graph may not appear in some environments.

These plot types (**Line Plot, Bar Chart, Horizontal Bar Chart, Pie Chart, Histogram, Scatter Plot, Subplots, Grid, Legends, Figure Size, Save Figure, and Pie Chart Explode**) cover more than **90% of Matplotlib usage** in beginner-to-intermediate **Data Science, Pandas, Analytics, and Machine Learning** projects.

Link : CHATGPT

`plt.figure(figsize=(8,5))` means:

`plt.figure(figsize=(8,5))`

Breakdown

- **figure()** → Creates a new graph window (canvas).
- **figsize** → Sets the size of the graph.
- **(8,5)** → (width, height)

So:

`figsize=(8,5)`

means:

Value	Meaning
-------	---------

8	Width = 8 inches
---	------------------

5	Height = 5 inches
---	-------------------

Visual Idea

Width = 8 inches

<----->

+-----+	
+-----+	

Height = 5 inches

Example

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))

plt.plot([1,2,3,4], [10,20,15,30])

plt.title("Sales Data")
plt.show()
```

Compare Different Sizes

Small Figure

`plt.figure(figsize=(4,3))`

Output: Small graph

Medium Figure

```
plt.figure(figsize=(8,5))
```

Output: Normal graph

Large Figure

```
plt.figure(figsize=(12,6))
```

Output: Wide graph

Why Use `figsize`?

1. Make graphs easier to read.
2. Prevent labels from overlapping.
3. Create graphs suitable for reports and presentations.
4. Control the width and height of the chart.

Remember

```
plt.figure(figsize=(width, height))
```

Example:

```
plt.figure(figsize=(10,6))
```

- Width = 10 inches
- Height = 6 inches

This line **must be written before** `plt.plot()`, `plt.bar()`, **etc.**, otherwise the size may not change for that figure.

Link : CHATGPT

`plt.subplot(1,2,1)` is used to **divide one figure into multiple plots**.

Syntax

```
plt.subplot(rows, columns, position)
```

Example

```
plt.subplot(1,2,1)
```

means:

Value	Meaning
1	Total rows
2	Total columns
1	Position number

Visual Representation

```
+-----+-----+
| Plot 1 | Plot 2 |
| Position 1 | Position 2 |
+-----+-----+
```

Here:

```
plt.subplot(1,2,1)
```

Selects **Plot 1** (left side).

And

```
plt.subplot(1,2,2)
```

Selects **Plot 2** (right side).

Complete Example

```
import matplotlib.pyplot as plt

x = [1,2,3,4]

plt.subplot(1,2,1)
plt.plot(x,[1,4,9,16])
plt.title("Square")

plt.subplot(1,2,2)
plt.plot(x,[1,2,3,4])
plt.title("Linear")
```

```
plt.show()
```

Output:

```
+-----+-----+
|      Square      |      Linear      |
|      /           |      /           |
|      /           |      /           |
+-----+-----+
```

Another Example

```
plt.subplot(2,2,1)
```

means:

- 2 rows
- 2 columns
- Position 1

Layout:

```
+-----+-----+
|      1      |      2      |
+-----+-----+
|      3      |      4      |
+-----+-----+
```

Positions

```
plt.subplot(2,2,1)  # Top Left
plt.subplot(2,2,2)  # Top Right
plt.subplot(2,2,3)  # Bottom Left
plt.subplot(2,2,4)  # Bottom Right
```

Easy Formula

```
plt.subplot(rows, columns, position)
```

Think:

How many rows?
How many columns?
Which box do I want to draw in?

Example:

```
plt.subplot(2,3,5)
```

means:

2 rows

3 columns

```
+---+---+---+
| 1 | 2 | 3 |
+---+---+---+
| 4 | 5 | 6 |
+---+---+---+
```

Current plot = Position 5

Shortcut to Remember

```
plt.subplot(1,2,1)
```

= 1 row, 2 columns, first plot

```
plt.subplot(1,2,2)
```

= 1 row, 2 columns, second plot 

Link : [CHATGPT](#)

Explode in Pie Chart (Matplotlib)

The **explode** parameter is used to **separate one or more slices from the pie chart** to highlight them.

Syntax

```
plt.pie(  
    sizes,  
    labels=labels,  
    explode=explode  
)
```

Example 1: Explode One Slice

```
import matplotlib.pyplot as plt  
  
sizes = [40, 30, 20, 10]  
labels = ["Python", "Java", "PHP", "C"]  
  
explode = [0.1, 0, 0, 0]  
  
plt.pie(  
    sizes,  
    labels=labels,  
    explode=explode,  
    autopct="%1.1f%%"  
)  
  
plt.title("Programming Language Usage")  
plt.show()
```

Explanation

```
explode = [0.1, 0, 0, 0]
```

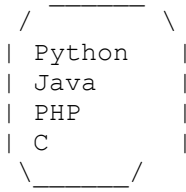
Slice Value

Python	0.1
Java	0
PHP	0
C	0

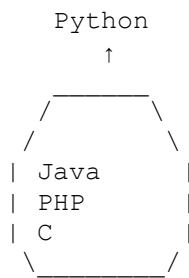
- Python slice moves outward.
 - Other slices remain in place.
-

Visual Idea

Without explode:



With explode:



Python slice is separated from the pie.

Example 2: Explode Multiple Slices

```
import matplotlib.pyplot as plt

sizes = [40, 30, 20, 10]
labels = ["Python", "Java", "PHP", "C"]

explode = [0.1, 0.2, 0, 0]

plt.pie(
    sizes,
    labels=labels,
    explode=explode,
    autopct="%1.1f%%"
)

plt.show()
```

Result

- Python moves out by 0.1
 - Java moves out by 0.2
 - PHP and C stay in place
-

Meaning of Values

Value	Meaning
0	No separation

Value	Meaning
0.1	Small separation
0.2	Medium separation
0.3	Large separation

Example:

```
explode = [0, 0, 0, 0]
```

No slice moves.

```
explode = [0.2, 0, 0, 0]
```

First slice moves outward.

Real-Life Use

Explode is used to highlight:

- Highest sales product
 - Most popular programming language
 - Largest market share
 - Top-performing department
-

Interview Question

Q: What is explode in a Pie Chart?

Answer:

The `explode` parameter is used to separate one or more slices from a pie chart to highlight important data.

Example:

```
explode = [0.1, 0, 0, 0]
```

This moves the first slice slightly away from the center of the pie chart.

Link : [CHATGPT](#)

Link : [ALL](#)

